SA UVM Guide

An annotated, source-complete walkthrough of the SA serial adder verification environment

Project baseline: sa_VHDL_UVM_all_transaction_methods.zip | RTL: VHDL-2008 | Testbench: SystemVerilog / UVM | Questa mixed-language workflow

How to use this guide

Read Chapters 1–5 sequentially to understand the testbench stimulus chain. Chapters 6–9 explain the observers and verification infrastructure. Appendix A reproduces every textual source file from the supplied project for cross-reference. No simulator execution is claimed.

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1. Understanding the SA DUT

The SA design is a serial adder. Each request consists of two 2-bit operands (ina and inb). Its VHDL ports are single-bit ina and inb because the operands are transmitted serially, most significant bit first. A three-bit sum is emitted serially as result_o and qualified by en_o.

Signal-level connection: sa_sequence → sa_sequencer → sa_driver → sa_if → sa (VHDL DUT); sa input/output monitors → expected/actual analysis streams → sa_scoreboard.

- clk: DUT rising-edge sample clock; driver prepares values at falling edges.
- en_i: request initiation; asserted during first operand bit in the supplied driver.
- ina/inb: serial operand pins; bit[1] is presented before bit[0].
- result_o: output result pin, replacing the reserved VHDL identifier out.
- en_o: output qualification pulse; monitor behavior determines exactly which cycles become transactions.

2. Transaction: the unit of stimulus

A UVM transaction is a SystemVerilog class that represents one meaningful transfer independently of pin-level timing. sa_transaction extends uvm_sequence_item. rand bit[1:0] ina and inb are randomized operand values; bit[2:0] out holds the sum. The transaction does not control en_i or clock edges: those belong to the driver.

The uvm_object_utils macro registers this type with the UVM factory. type_id::create allows future factory overrides. new constructs the object. The new revision implements convert2string, do_copy, do_compare, do_print, do_record, do_pack, and do_unpack; all are exercised from the testbench. In this source, do_compare compares all three fields, although the scoreboard intentionally compares the result field only.

sa_transaction.svh

Study the class declaration and methods below. A transaction object can carry both generated inputs and observed/predicted result values, but each producer must populate the fields it owns.

Complete source: sa_transaction.svh (78 lines)

```
`ifndef __SA_TRANSACTION_SVH__
`define __SA_TRANSACTION_SVH__
// One 2-bit input pair and its 3-bit sum; explicit UVM field hooks.
class sa_transaction extends uvm_sequence_item;
  `uvm_object_utils(sa_transaction)
  rand bit [1:0] ina;
  rand bit [1:0] inb;
       bit [2:0] out;

  function new(string name = "sa_transaction");
    super.new(name);
  endfunction

  function string convert2string();
    return $sformatf("ina=%0d (0x%0h) inb=%0d (0x%0h) out=%0d (0x%0h)",
                     ina, ina, inb, inb, out, out);
  endfunction

  // Invoked via destination.copy(source).
  function void do_copy(uvm_object rhs);
    sa_transaction src;
    if (!$cast(src, rhs))
      `uvm_fatal("SA_COPY_TYPE", "rhs is not an sa_transaction")
    super.do_copy(rhs);
    ina = src.ina;
    inb = src.inb;
    out = src.out;
  endfunction

  // Invoked via expected.compare(actual, comparer).
  function bit do_compare(uvm_object rhs, uvm_comparer comparer);
    sa_transaction other;
    bit matches;
    if (!$cast(other, rhs)) begin
      `uvm_error("SA_COMPARE_TYPE", "rhs is not an sa_transaction")
      return 0;
    end
    matches = super.do_compare(rhs, comparer);
    matches &= comparer.compare_field("ina", ina, other.ina, 2, UVM_DEC);
    matches &= comparer.compare_field("inb", inb, other.inb, 2, UVM_DEC);
    matches &= comparer.compare_field("out", out, other.out, 3, UVM_DEC);
    return matches;
  endfunction

  // Invoked via tx.print() / tx.sprint().
  function void do_print(uvm_printer printer);
    super.do_print(printer);
    printer.print_field_int("ina", ina, 2, UVM_DEC);
    printer.print_field_int("inb", inb, 2, UVM_DEC);
    printer.print_field_int("out", out, 3, UVM_DEC);
  endfunction
```

```
  // Invoked via tx.record(recorder), normally after a stream is opened.
  function void do_record(uvm_recorder recorder);
    super.do_record(recorder);
    recorder.record_field("ina", ina, 2, UVM_DEC);
    recorder.record_field("inb", inb, 2, UVM_DEC);
    recorder.record_field("out", out, 3, UVM_DEC);
  endfunction

  // Invoked via pack_bits(), pack_bytes(), or pack_ints().
  // Width/order MUST match do_unpack: 2 + 2 + 3 = 7 bits.
  function void do_pack(uvm_packer packer);
    super.do_pack(packer);
    packer.pack_field_int(ina, 2);
    packer.pack_field_int(inb, 2);
    packer.pack_field_int(out, 3);
  endfunction

  // Invoked via unpack_bits(), unpack_bytes(), or unpack_ints().
  function void do_unpack(uvm_packer packer);
    super.do_unpack(packer);
    ina = packer.unpack_field_int(2);
    inb = packer.unpack_field_int(2);
    out = packer.unpack_field_int(3);
  endfunction
endclass : sa_transaction
`endif
```

2A. Seven transaction methods: implementation and actual exercise

This updated revision corresponds to sa_VHDL_UVM_all_transaction_methods.zip. The UVM user-facing APIs call virtual hook methods on sa_transaction. Overriding the hook defines field semantics; invoking the public API from the test or scoreboard causes the override to execute. This distinction is important: do_copy is not the same API as copy().

Transaction hook	Called through	What happens
convert2string()	tx.convert2string()	Return an operand/output summary
do_copy(rhs)	dst.copy(src)	Transfer fields after safe type cast
do_compare(rhs, comparer)	lhs.compare(rhs, cmp)	Check corresponding transaction fields
do_print(printer)	tx.sprint()	Print via UVM printer framework
do_record(recorder)	tx.record(rec)	Record fields; database setup is simulator-specific
do_pack(packer)	tx.pack_bits(bits)	Serialize 2+2+3 = 7 bits
do_unpack(packer)	tx.unpack_bits(bits)	Deserialize the seven bits

2A.1 The transaction class: override contracts

In sa_transaction.svh, every method calls its parent where appropriate and processes ina (2 bits), inb (2 bits), and out (3 bits). do_copy uses $cast to verify rhs is a compatible transaction. do_compare must return a bit result; comparisons should use the passed comparer for consistent reporting. The packer/unpacker must use the same field order and widths.

```
`ifndef __SA_TRANSACTION_SVH__
`define __SA_TRANSACTION_SVH__
// One 2-bit input pair and its 3-bit sum; explicit UVM field hooks.
class sa_transaction extends uvm_sequence_item;
  `uvm_object_utils(sa_transaction)
  rand bit [1:0] ina;
```

```
  rand bit [1:0] inb;
       bit [2:0] out;

  function new(string name = "sa_transaction");
    super.new(name);
  endfunction

  function string convert2string();
    return $sformatf("ina=%0d (0x%0h) inb=%0d (0x%0h) out=%0d (0x%0h)",
                     ina, ina, inb, inb, out, out);
  endfunction

  // Invoked via destination.copy(source).
  function void do_copy(uvm_object rhs);
    sa_transaction src;
    if (!$cast(src, rhs))
      `uvm_fatal("SA_COPY_TYPE", "rhs is not an sa_transaction")
    super.do_copy(rhs);
    ina = src.ina;
    inb = src.inb;
    out = src.out;
  endfunction

  // Invoked via expected.compare(actual, comparer).
  function bit do_compare(uvm_object rhs, uvm_comparer comparer);
    sa_transaction other;
    bit matches;
    if (!$cast(other, rhs)) begin
      `uvm_error("SA_COMPARE_TYPE", "rhs is not an sa_transaction")
      return 0;
    end
    matches = super.do_compare(rhs, comparer);
    matches &= comparer.compare_field("ina", ina, other.ina, 2, UVM_DEC);
    matches &= comparer.compare_field("inb", inb, other.inb, 2, UVM_DEC);
    matches &= comparer.compare_field("out", out, other.out, 3, UVM_DEC);
    return matches;
  endfunction

  // Invoked via tx.print() / tx.sprint().
  function void do_print(uvm_printer printer);
    super.do_print(printer);
    printer.print_field_int("ina", ina, 2, UVM_DEC);
    printer.print_field_int("inb", inb, 2, UVM_DEC);
    printer.print_field_int("out", out, 3, UVM_DEC);
  endfunction

  // Invoked via tx.record(recorder), normally after a stream is opened.
  function void do_record(uvm_recorder recorder);
    super.do_record(recorder);
    recorder.record_field("ina", ina, 2, UVM_DEC);
    recorder.record_field("inb", inb, 2, UVM_DEC);
    recorder.record_field("out", out, 3, UVM_DEC);
  endfunction

  // Invoked via pack_bits(), pack_bytes(), or pack_ints().
  // Width/order MUST match do_unpack: 2 + 2 + 3 = 7 bits.
  function void do_pack(uvm_packer packer);
    super.do_pack(packer);
    packer.pack_field_int(ina, 2);
    packer.pack_field_int(inb, 2);
    packer.pack_field_int(out, 3);
  endfunction

  // Invoked via unpack_bits(), unpack_bytes(), or unpack_ints().
  function void do_unpack(uvm_packer packer);
    super.do_unpack(packer);
    ina = packer.unpack_field_int(2);
    inb = packer.unpack_field_int(2);
    out = packer.unpack_field_int(3);
  endfunction
endclass : sa_transaction
```

```
`endif
```

2A.2 How the test exercises the public APIs

sa_test.exercise_transaction_methods() runs before the random DUT sequence. It constructs original, copied, decoded and deliberately changed objects. The following is the exact source of the test method:

```
  task exercise_transaction_methods();
    sa_transaction original, copied, decoded, changed;
    uvm_comparer cmp;
    uvm_recorder rec;
    bit packed_bits[];
    int nbits;

    original = sa_transaction::type_id::create("original");
    copied   = sa_transaction::type_id::create("copied");
    decoded  = sa_transaction::type_id::create("decoded");
    changed  = sa_transaction::type_id::create("changed");
    original.ina = 2'd2;
    original.inb = 2'd3;
    original.out = 3'd5;

    // 1. convert2string()
    `uvm_info("SA_METHOD_STRING", original.convert2string(), UVM_LOW)

    // 2. do_copy() through copy()
    copied.copy(original);
    // 3. do_compare() through compare()
    cmp = new();
    if (!original.compare(copied, cmp))
      `uvm_fatal("SA_METHOD_COPY", "copy/compare failed")

    // 4. do_print() through sprint()/print()
    `uvm_info("SA_METHOD_PRINT", original.sprint(), UVM_LOW)

    // 5. do_record() through record() with a recorder.
    // Backend-specific waveform/database output needs the simulator's
    // transaction-recording configuration.
    rec = new("sa_method_recorder");
    original.record(rec);
    `uvm_info("SA_METHOD_RECORD", "Called record(rec): do_record executed", UVM_LOW)

    // 6. do_pack() and 7. do_unpack() through the bit-array APIs.
    nbits = original.pack_bits(packed_bits);
    if (nbits != 7)
      `uvm_fatal("SA_METHOD_PACK", $sformatf("Expected 7 packed bits; got %0d", nbits))
    void'(decoded.unpack_bits(packed_bits));
    cmp = new();
    if (!original.compare(decoded, cmp))
      `uvm_fatal("SA_METHOD_UNPACK", "pack/unpack round-trip failed")

    // Negative compare demonstrates field-by-field mismatch diagnostics.
    changed.copy(original);
    changed.out = original.out ^ 3'b001;
    cmp = new();
    cmp.show_max = 0; // Suppress intentional mismatch diagnostic output.
    if (original.compare(changed, cmp))
      `uvm_fatal("SA_METHOD_NEGATIVE", "Deliberately changed output compared equal")
    `uvm_info("SA_METHODS_PASS", "All seven UVM transaction methods exercised", UVM_NONE)
  endtask
```

2A.3 Expected checks and debugging

- Copy and equality: copied.copy(original) followed by original.compare(copied,cmp) must pass.
- Print and logging: sprint() returns detailed printer output while convert2string() returns the compact custom representation.
- Recording: record(rec) exercises the hook, but a visible transaction database requires appropriate Questa/UVM recording setup.

- Round trip: pack_bits() must report seven bits and unpack_bits() must reproduce original fields before compare().
- Negative comparison: changed.out intentionally differs; a false comparison is success for the negative test.
- The scoreboard also uses copy() and compare() for actual-vs-expected normalization. This assumes input/output transactions are paired in order.
- The test emits SA_METHODS_PASS when method checks complete, while ADDER_SUMMARY reports normal DUT results.

2A.4 Where to inspect in the source

Read sa_transaction.svh for all implementations, sa_test.sv for explicit method invocations, sa_monitor.sv for convert2string() during observation, and sa_scoreboard.sv for copy/compare during checking. These files are reproduced at their regular chapter locations. README_UVM_METHODS.md summarizes the usage and validation limitations.

3. Sequence and sequencer

A sequence selects and orders stimulus. sa_sequence extends uvm_sequence#(sa_transaction) and generates 15 requests by default. Each loop constructs a fresh item, calls start_item, randomizes it, and calls finish_item. The sequencer arbitrates sequence items and exposes the sequence-item export consumed by the driver.

The typedef sa_sequencer = uvm_sequencer#(sa_transaction) uses the stock parameterized sequencer; a bespoke class is unnecessary for simple arbitration.

sa_sequencer.sv

Note the distinction: sa_sequence is a UVM object that executes body(), while sa_sequencer is a UVM component used as a communication bridge. The test starts the sequence on the agent sequencer.

Complete source: sa_sequencer.sv (24 lines)

```
class sa_sequence extends uvm_sequence#(sa_transaction);
 `uvm_object_utils(sa_sequence)
 extern function   new(string name = "");
 extern task    body();
endclass: sa_sequence
typedef uvm_sequencer#(sa_transaction) sa_sequencer;
function sa_sequence::new(string name = "");
 super.new(name);
endfunction: new
task sa_sequence::body();
 sa_transaction sa_tx;
 if (starting_phase != null)
     starting_phase.raise_objection(this);
 repeat(15) begin
  sa_tx = sa_transaction::type_id::create(.name("sa_tx") ,
               .contxt(get_full_name()) );
  start_item(sa_tx);     /* handshake with driver::get function */
  if( !sa_tx.randomize() )
   `uvm_error("", "failed to randomize...");
  finish_item(sa_tx);    /* handshake with driver::get function */
 end
 if (starting_phase != null)
  starting_phase.drop_objection(this);
endtask: body
```

4. Driver: translating transactions into clocked pins

sa_driver extends uvm_driver#(sa_transaction), obtains a virtual sa_if via its configuration object, and repeatedly asks the sequencer for the next transaction. Its job is stimulus generation only, not result checking.

Clock discipline: the DUT samples on rising edges. The driver changes pins on falling edges to reduce active-region races. On cycle 0 it asserts en_i and presents each operand MSB; on cycle 1 it presents each LSB. Remaining cycles hold inactive operand pins while the serial result is produced. After the sixth cycle it acknowledges completion with item_done().

sa_driver.sv

Key APIs: seq_item_port.get_next_item(req) blocks until an item is available; @(negedge m_vif.sig_clock) aligns stimulus before the DUT sampling edge; item_done() completes the request handshake. Null interface or missing configuration produce uvm_fatal.

Complete source: sa_driver.sv (34 lines)

```
class sa_driver extends uvm_driver #(sa_transaction);
  `uvm_component_utils(sa_driver)
  virtual sa_if m_vif;
  clk_data_config drv_config;
  function new(string name, uvm_component parent); super.new(name,parent); endfunction
  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    if (!uvm_config_db#(clk_data_config)::get(this,"","config",drv_config))
      `uvm_fatal("DRIVER_CFG","Missing clk_data_config")
    m_vif = drv_config.m_vif;
    if (m_vif == null) `uvm_fatal("DRIVER_VIF","Null virtual interface")
  endfunction
  task run_phase(uvm_phase phase);
    sa_transaction req;
    m_vif.sig_en_i = 0;
    m_vif.sig_ina = 0;
    m_vif.sig_inb = 0;
    forever begin
      seq_item_port.get_next_item(req);
      // DUT consumes at rising edges; present stimulus at falling edges.
      for (int cycle=0; cycle<6; cycle++) begin
        @(negedge m_vif.sig_clock);
        m_vif.sig_en_i <= (cycle == 0);
        m_vif.sig_ina  <= (cycle == 0) ? req.ina[1] :
                          (cycle == 1) ? req.ina[0] : 1'b0;
        m_vif.sig_inb  <= (cycle == 0) ? req.inb[1] :
                          (cycle == 1) ? req.inb[0] : 1'b0;
      end
      // At posedge after cycle 5 the RTL completes its six-cycle operation.
      @(posedge m_vif.sig_clock);
      seq_item_port.item_done();
    end
  endtask
endclass
```

5. Complete stimulus handshake

Sequence calls start_item → sequencer grants transaction → sequence randomizes and calls finish_item → driver get_next_item → driver presents pins on falling edges → VHDL DUT captures pins on rising edges → driver item_done → sequence may advance. The monitor independently observes pins; a scoreboard independently checks outcomes.

Example: ina=2'b11 (3), inb=2'b10 (2): the mathematical expected output is 3'b101 (5). Use waveforms to confirm the precise serial output sampling order and en_o qualification in this implementation.

6. Monitors and prediction

A UVM monitor converts interface activity back into transaction objects and publishes them through analysis ports. This project has explicit input/expected and output/actual monitoring. Every published sample should own a separate object or immutable clone so later updates cannot alter an item already queued downstream.

sa_monitor.sv

Read this source alongside the RTL timing. Check transaction initialization, sampling cycles, DPI argument widths and which side generates the expected result.

Complete source: sa_monitor.sv (97 lines)

```
// Independent agents: input monitor reports expected transactions;
// output monitor reconstructs serial DUT results.
import "DPI-C" function chandle cl_predict_new(string name);
import "DPI-C" function int cl_predict_set_values_handle(int x,int y,chandle handle);

class sa_input_monitor extends uvm_monitor;
  `uvm_component_utils(sa_input_monitor)
  virtual sa_if m_vif;
  clk_data_config mon_config;
  uvm_analysis_port #(sa_transaction) input_ap;
  chandle hPredict;
  sa_transaction coverage_tx;
  covergroup sa_cg;
    ina_cp: coverpoint coverage_tx.ina;
    inb_cp: coverpoint coverage_tx.inb;
    ab_cross: cross ina_cp,inb_cp;
  endgroup
  function new(string name,uvm_component parent);
    super.new(name,parent);
    input_ap = new("input_ap",this);
    sa_cg = new();
  endfunction
  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    if (!uvm_config_db#(clk_data_config)::get(this,"","config",mon_config))
      `uvm_fatal("INPUT_CFG","Missing clk_data_config")
    m_vif = mon_config.m_vif;
    if (m_vif == null) `uvm_fatal("INPUT_VIF","Null virtual interface")
    hPredict = cl_predict_new("serial_adder_predictor");
    if (hPredict == null) `uvm_fatal("DPI","Failed to allocate predictor")
  endfunction
  task run_phase(uvm_phase phase);
    sa_transaction tx;
    int state = 0;
    forever begin
      // Inputs were driven on negedge and are stable at this posedge.
      @(posedge m_vif.sig_clock);
      if (state == 0 && m_vif.sig_en_i) begin
        tx = sa_transaction::type_id::create("input_tx");
        tx.ina = {1'b0,m_vif.sig_ina};
        tx.inb = {1'b0,m_vif.sig_inb};
        state = 1;
      end else if (state == 1) begin
        tx.ina = {tx.ina[0],m_vif.sig_ina};
        tx.inb = {tx.inb[0],m_vif.sig_inb};
        tx.out = cl_predict_set_values_handle(int'(tx.ina),int'(tx.inb),hPredict);
        coverage_tx = tx;
        if (mon_config.coverage_enable) sa_cg.sample();
        // One unique transaction object is allocated per transfer.
        input_ap.write(tx);
        `uvm_info("INPUT_MON",tx.convert2string(),UVM_MEDIUM)
        tx = null;
        state = 0;
      end
    end
  endtask
endclass

class sa_output_monitor extends uvm_monitor;
  `uvm_component_utils(sa_output_monitor)
  virtual sa_if m_vif;
  clk_data_config mon_config;
  uvm_analysis_port #(sa_transaction) output_ap;
  function new(string name,uvm_component parent);
    super.new(name,parent);
```

```
    output_ap = new("output_ap",this);
  endfunction
  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    if (!uvm_config_db#(clk_data_config)::get(this,"","config",mon_config))
      `uvm_fatal("OUTPUT_CFG","Missing clk_data_config")
    m_vif = mon_config.m_vif;
    if (m_vif == null) `uvm_fatal("OUTPUT_VIF","Null virtual interface")
  endfunction
  task run_phase(uvm_phase phase);
    sa_transaction tx;
    int state = 0;
    forever begin
      // Sample at negedge: RTL posedge nonblocking updates are settled.
      @(negedge m_vif.sig_clock);
      if (state == 0 && m_vif.sig_en_o) begin
        tx = sa_transaction::type_id::create("output_tx");
        tx.out = {m_vif.sig_out,2'b0};
        state = 1;
      end else if (state == 1) begin
        tx.out[1] = m_vif.sig_out;
        state = 2;
      end else if (state == 2) begin
        tx.out[0] = m_vif.sig_out;
        output_ap.write(tx);
        `uvm_info("OUTPUT_MON",tx.convert2string(),UVM_MEDIUM)
        tx = null;
        state = 0;
      end
    end
  endtask
endclass
```

predict_class.cpp

Read this source alongside the RTL timing. Check transaction initialization, sampling cycles, DPI argument widths and which side generates the expected result.

Complete source: predict_class.cpp (34 lines)

```
#include "vpi_user.h"       // use vpi_printf so all output goes to transcript
#include "predict.hpp"
#include <string>
using namespace std;

class   CL_PREDICT {
  int ina, inb;
  const string m_name;
public:
    CL_PREDICT(string name) : m_name(name) {}
    void  set_values (int,int);
    int   do_predict () {return ina+inb;}
    const char* name () {return m_name.c_str();}
};

void CL_PREDICT::set_values (int x, int y) {
    ina   = x;
    inb   = y;
}



void* cl_predict_new(const char* name) {
    return (void*)( new CL_PREDICT(name) );
}

int cl_predict_set_values_handle(int x, int y, void* handle) {
    CL_PREDICT* r = static_cast<CL_PREDICT *>(handle);
    r->set_values(x,y);
    int out = r->do_predict();
```

```
    //vpi_printf("%s c++ predict: %d\n", r->name(), out);
    return out;
}
```

predict.hpp

Read this source alongside the RTL timing. Check transaction initialization, sampling cycles, DPI argument widths and which side generates the expected result.

Complete source: predict.hpp (163 lines)

```
/* MTI_DPI */

/*
 * Copyright 2002-2016 Mentor Graphics Corporation.
 *
 * Note:
 *   This file is automatically generated.
 *   Please do not edit this file - you will lose your edits.
 *
 * Settings when this file was generated:
 *   PLATFORM = 'win64'
 */
#ifndef INCLUDED_PREDICT
#define INCLUDED_PREDICT

#ifdef __cplusplus
#define DPI_LINK_DECL  extern "C"
#else
#define DPI_LINK_DECL
#endif

#include "svdpi.h"


#ifndef MTI_INCLUDED_TYPEDEF_uvm_hdl_path_slice
#define MTI_INCLUDED_TYPEDEF_uvm_hdl_path_slice

typedef struct {
    const char* path;
    int offset;
    int size;
}  uvm_hdl_path_slice;

#endif

#ifndef MTI_INCLUDED_TYPEDEF_uvm_printer_row_info
#define MTI_INCLUDED_TYPEDEF_uvm_printer_row_info

typedef struct {
    int level;
    const char* name;
    const char* type_name;
    const char* size;
    const char* val;
}  uvm_printer_row_info;

#endif

#ifndef MTI_INCLUDED_TYPEDEF_uvm_reg_bus_op
#define MTI_INCLUDED_TYPEDEF_uvm_reg_bus_op

typedef struct {
    int kind;
    svBitVecVal addr[SV_PACKED_DATA_NELEMS(64)];
    svBitVecVal data[SV_PACKED_DATA_NELEMS(64)];
    int n_bits;
    svBitVecVal byte_en[SV_PACKED_DATA_NELEMS(8)];
    int status;
}  uvm_reg_bus_op;
```

```
#endif


DPI_LINK_DECL DPI_DLLESPEC
void*
cl_predict_new(
    const char* name);

DPI_LINK_DECL DPI_DLLESPEC
int
cl_predict_set_values_handle(
    int x,
    int y,
    void* handle);

DPI_LINK_DECL DPI_DLLESPEC
const char*
uvm_dpi_get_next_arg_c(
    int init);

DPI_LINK_DECL DPI_DLLESPEC
const char*
uvm_dpi_get_tool_name_c();

DPI_LINK_DECL DPI_DLLESPEC
const char*
uvm_dpi_get_tool_version_c();

DPI_LINK_DECL DPI_DLLESPEC
void*
uvm_dpi_regcomp(
    const char* regex);

DPI_LINK_DECL DPI_DLLESPEC
int
uvm_dpi_regexec(
    void* preg,
    const char* str);

DPI_LINK_DECL DPI_DLLESPEC
void
uvm_dpi_regfree(
    void* preg);

DPI_LINK_DECL DPI_DLLESPEC
void
uvm_dump_re_cache();

DPI_LINK_DECL DPI_DLLESPEC
const char*
uvm_glob_to_re(
    const char* glob);

DPI_LINK_DECL DPI_DLLESPEC
int
uvm_hdl_check_path(
    const char* path);

DPI_LINK_DECL DPI_DLLESPEC
int
uvm_hdl_deposit(
    const char* path,
    const svLogicVecVal* value);

DPI_LINK_DECL DPI_DLLESPEC
int
uvm_hdl_force(
    const char* path,
    const svLogicVecVal* value);
```

```
DPI_LINK_DECL DPI_DLLESPEC
int
uvm_hdl_read(
    const char* path,
    svLogicVecVal* value);

DPI_LINK_DECL DPI_DLLESPEC
int
uvm_hdl_release(
    const char* path);

DPI_LINK_DECL DPI_DLLESPEC
int
uvm_hdl_release_and_read(
    const char* path,
    svLogicVecVal* value);

DPI_LINK_DECL DPI_DLLESPEC
int
uvm_re_match(
    const char* re,
    const char* str);

DPI_LINK_DECL void
m__uvm_report_dpi(
    int severity,
    const char* id,
    const char* message,
    int verbosity,
    const char* filename,
    int line);

#endif
```

7. Scoreboard and test control

The scoreboard receives expected and actual transactions through analysis exports feeding two analysis FIFOs. run_phase retrieves one item from each FIFO, compares the out fields with 4-state case equality, emits uvm_info on pass or uvm_error on mismatch, and increments counters. check_phase identifies missing comparisons and FIFO leftovers; report_phase prints counts.

The test raises a run-phase objection, starts the sequence on its sequencer, then allows output latency to drain before dropping the objection. A robust regression should explicitly check that the number of comparisons equals the number of stimulated transactions, and that zero failures occurred.

sa_scoreboard.sv

Inspect the phase behavior carefully: run_phase is automatically scheduled by UVM, whereas an unrelated task run() is not.

Complete source: sa_scoreboard.sv (54 lines)

```
class sa_scoreboard extends uvm_scoreboard;
  `uvm_component_utils(sa_scoreboard)
  uvm_analysis_export #(sa_transaction) expected_export;
  uvm_analysis_export #(sa_transaction) actual_export;
  uvm_tlm_analysis_fifo #(sa_transaction) expected_fifo;
  uvm_tlm_analysis_fifo #(sa_transaction) actual_fifo;
  int unsigned pass_count, fail_count;
  function new(string name,uvm_component parent);super.new(name,parent);endfunction
  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    expected_export = new("expected_export",this);
    actual_export = new("actual_export",this);
    expected_fifo = new("expected_fifo",this);
    actual_fifo = new("actual_fifo",this);
  endfunction
```

```
  function void connect_phase(uvm_phase phase);
    super.connect_phase(phase);
    expected_export.connect(expected_fifo.analysis_export);
    actual_export.connect(actual_fifo.analysis_export);
  endfunction
  task run_phase(uvm_phase phase);
    sa_transaction expected,actual,actual_normalized;
    uvm_comparer comparer;
    forever begin
      expected_fifo.get(expected);
      actual_fifo.get(actual);
      // Output monitor observes only the serial sum. Supply the associated
      // input operands from the expected transaction so compare() checks
      // all three sa_transaction fields without false input mismatches.
      actual_normalized = sa_transaction::type_id::create("actual_normalized");
      actual_normalized.copy(actual);
      actual_normalized.ina = expected.ina;
      actual_normalized.inb = expected.inb;
      comparer = new();
      if (expected.compare(actual_normalized, comparer)) begin
        pass_count++;
        `uvm_info("ADDER_PASS",$sformatf("A=%0d B=%0d expected=%0d
actual=%0d",expected.ina,expected.inb,expected.out,actual.out),UVM_LOW)
      end else begin
        fail_count++;
        `uvm_error("ADDER_MISMATCH",$sformatf("A=%0d B=%0d expected=%0d
actual=%0d",expected.ina,expected.inb,expected.out,actual.out))
      end
    end
  endtask
  function void check_phase(uvm_phase phase);
    super.check_phase(phase);
    if (pass_count+fail_count==0) `uvm_error("NO_CHECKS","No transactions compared")
    if (expected_fifo.used()!=0 || actual_fifo.used()!=0)
      `uvm_error("UNMATCHED",$sformatf("Remaining expected=%0d
actual=%0d",expected_fifo.used(),actual_fifo.used()))
  endfunction
  function void report_phase(uvm_phase phase);
    super.report_phase(phase);
    `uvm_info("ADDER_SUMMARY",$sformatf("PASSED=%0d FAILED=%0d",pass_count,fail_count),UVM_NONE)
  endfunction
endclass
```

sa_test.sv

Inspect the phase behavior carefully: run_phase is automatically scheduled by UVM, whereas an unrelated task run() is not.

Complete source: sa_test.sv (75 lines)

```
class sa_test extends uvm_test;
  `uvm_component_utils(sa_test)
  sa_env sa_env;
  function new(string name,uvm_component parent);super.new(name,parent);endfunction
  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    sa_env=sa_env::type_id::create("sa_env",this);
  endfunction
  // This method is run before the DUT sequence and exercises every requested
  // uvm_object hook through its standard PUBLIC API.
  task exercise_transaction_methods();
    sa_transaction original, copied, decoded, changed;
    uvm_comparer cmp;
    uvm_recorder rec;
    bit packed_bits[];
    int nbits;

    original = sa_transaction::type_id::create("original");
    copied   = sa_transaction::type_id::create("copied");
    decoded  = sa_transaction::type_id::create("decoded");
    changed  = sa_transaction::type_id::create("changed");
    original.ina = 2'd2;
    original.inb = 2'd3;
```

```
    original.out = 3'd5;

    // 1. convert2string()
    `uvm_info("SA_METHOD_STRING", original.convert2string(), UVM_LOW)

    // 2. do_copy() through copy()
    copied.copy(original);
    // 3. do_compare() through compare()
    cmp = new();
    if (!original.compare(copied, cmp))
      `uvm_fatal("SA_METHOD_COPY", "copy/compare failed")

    // 4. do_print() through sprint()/print()
    `uvm_info("SA_METHOD_PRINT", original.sprint(), UVM_LOW)

    // 5. do_record() through record() with a recorder.
    // Backend-specific waveform/database output needs the simulator's
    // transaction-recording configuration.
    rec = new("sa_method_recorder");
    original.record(rec);
    `uvm_info("SA_METHOD_RECORD", "Called record(rec): do_record executed", UVM_LOW)

    // 6. do_pack() and 7. do_unpack() through the bit-array APIs.
    nbits = original.pack_bits(packed_bits);
    if (nbits != 7)
      `uvm_fatal("SA_METHOD_PACK", $sformatf("Expected 7 packed bits; got %0d", nbits))
    void'(decoded.unpack_bits(packed_bits));
    cmp = new();
    if (!original.compare(decoded, cmp))
      `uvm_fatal("SA_METHOD_UNPACK", "pack/unpack round-trip failed")

    // Negative compare demonstrates field-by-field mismatch diagnostics.
    changed.copy(original);
    changed.out = original.out ^ 3'b001;
    cmp = new();
    cmp.show_max = 0; // Suppress intentional mismatch diagnostic output.
    if (original.compare(changed, cmp))
      `uvm_fatal("SA_METHOD_NEGATIVE", "Deliberately changed output compared equal")
    `uvm_info("SA_METHODS_PASS", "All seven UVM transaction methods exercised", UVM_NONE)
  endtask

  task run_phase(uvm_phase phase);
    sa_sequence seq;
    phase.raise_objection(this);
    exercise_transaction_methods();
    seq=sa_sequence::type_id::create("seq");
    seq.start(sa_env.sa_agent.sa_seqr);
    // Wait until the final three-bit output can be observed and compared.
    repeat (8) @(posedge sa_env.sa_agent.agent_config.m_vif.sig_clock);
    phase.drop_objection(this);
  endtask
endclass
```

8. Configuration, factory and environment

UVM constructs components in build_phase, obtains configuration using uvm_config_db, and connects TLM ports in connect_phase. The virtual interface bridges class-based testbench code to the static sa_if instance in the top-level module. The factory enables derived sequences/drivers to replace defaults without hard-coding constructors.

sa_if.sv

This is infrastructure source: trace each create/get/set/connect and identify where the interface handle is first bound.

Complete source: sa_if.sv (11 lines)

```
`ifndef __SA_IF_SV__
```

```
`define __SA_IF_SV__
interface sa_if;
 logic  sig_clock;
 logic  sig_ina;
 logic  sig_inb;
 logic  sig_en_i;
 logic  sig_out;
 logic  sig_en_o;
endinterface: sa_if
`endif /* __SA_IF_SV__ */
```

sa_config.sv

This is infrastructure source: trace each create/get/set/connect and identify where the interface handle is first bound.

Complete source: sa_config.sv (15 lines)

```
class sa_config extends uvm_object;
  `uvm_object_utils(sa_config)
  virtual sa_if m_vif;
  uvm_active_passive_enum is_active_sa = UVM_ACTIVE;
  bit checks_enable_sa = 1;
  bit coverage_enable_sa = 1;
  function new(string name="sa_config"); super.new(name);endfunction
endclass
class clk_data_config extends uvm_object;
  `uvm_object_utils(clk_data_config)
  virtual sa_if m_vif;
  uvm_active_passive_enum is_active_sa=UVM_ACTIVE;
  bit coverage_enable=1;
  function new(string name="clk_data_config"); super.new(name);endfunction
endclass
```

clkndata_config.sv

This is infrastructure source: trace each create/get/set/connect and identify where the interface handle is first bound.

Complete source: clkndata_config.sv (14 lines)

```
`ifndef CLKNDATA_CONFIG_SV
`define CLKNDATA_CONFIG_SV
class clkndata_config extends uvm_object;
  // Do not register config class with the factory
  virtual clkndata_if      vif;
  uvm_active_passive_enum  is_active = UVM_ACTIVE;
  bit                      coverage_enable;
  bit                      checks_enable;
  extern function new(string name = "");
endclass : clkndata_config
function clkndata_config::new(string name = "");
  super.new(name);
endfunction : new
`endif // CLKNDATA_CONFIG_SV
```

sa_agent.sv

This is infrastructure source: trace each create/get/set/connect and identify where the interface handle is first bound.

Complete source: sa_agent.sv (41 lines)

```
class sa_agent extends uvm_agent;
  `uvm_component_utils(sa_agent)
  sa_config agent_config;
  sa_sequencer sa_seqr;
  sa_driver sa_drvr;
  sa_input_monitor input_mon;
  sa_output_monitor output_mon;
  uvm_analysis_port #(sa_transaction) expected_ap;
  uvm_analysis_port #(sa_transaction) actual_ap;
  function new(string name,uvm_component parent);
```

```
    super.new(name,parent);
    expected_ap=new("expected_ap",this);
    actual_ap=new("actual_ap",this);
  endfunction
  function void build_phase(uvm_phase phase);
    clk_data_config cfg;
    super.build_phase(phase);
    if (!uvm_config_db#(sa_config)::get(this,"","config",agent_config))
      `uvm_fatal("AGENT_CFG","Missing sa_config")
    if (agent_config.m_vif==null) `uvm_fatal("AGENT_VIF","Null vif")
    cfg=new("cfg");
    cfg.m_vif=agent_config.m_vif;
    cfg.is_active_sa=agent_config.is_active_sa;
    cfg.coverage_enable=agent_config.coverage_enable_sa;
    uvm_config_db#(clk_data_config)::set(this,"input_mon","config",cfg);
    uvm_config_db#(clk_data_config)::set(this,"output_mon","config",cfg);
    input_mon=sa_input_monitor::type_id::create("input_mon",this);
    output_mon=sa_output_monitor::type_id::create("output_mon",this);
    if (agent_config.is_active_sa==UVM_ACTIVE) begin
      uvm_config_db#(clk_data_config)::set(this,"sa_drvr","config",cfg);
      sa_seqr=sa_sequencer::type_id::create("sa_seqr",this);
      sa_drvr=sa_driver::type_id::create("sa_drvr",this);
    end
  endfunction
  function void connect_phase(uvm_phase phase);
    super.connect_phase(phase);
    input_mon.input_ap.connect(expected_ap);
    output_mon.output_ap.connect(actual_ap);
    if (sa_drvr!=null) sa_drvr.seq_item_port.connect(sa_seqr.seq_item_export);
  endfunction
endclass
```

sa_env.sv

This is infrastructure source: trace each create/get/set/connect and identify where the interface handle is first bound.

Complete source: sa_env.sv (20 lines)

```
class sa_env extends uvm_env;
  `uvm_component_utils(sa_env)
  sa_config env_config;
  sa_agent sa_agent;
  sa_scoreboard sa_sb;
  function new(string name,uvm_component parent);super.new(name,parent);endfunction
  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    if (!uvm_config_db#(sa_config)::get(this,"","config",env_config))
      `uvm_fatal("ENV_CFG","Missing sa_config")
    uvm_config_db#(sa_config)::set(this,"sa_agent","config",env_config);
    sa_agent=sa_agent::type_id::create("sa_agent",this);
    sa_sb=sa_scoreboard::type_id::create("sa_sb",this);
  endfunction
  function void connect_phase(uvm_phase phase);
    super.connect_phase(phase);
    sa_agent.expected_ap.connect(sa_sb.expected_export);
    sa_agent.actual_ap.connect(sa_sb.actual_export);
  endfunction
endclass
```

sa_pkg.sv

This is infrastructure source: trace each create/get/set/connect and identify where the interface handle is first bound.

Complete source: sa_pkg.sv (16 lines)

```
`ifndef __SA_PKG_SV__
`define __SA_PKG_SV__
`include "uvm_macros.svh"
package sa_pkg;
 import uvm_pkg::*;
 `include "sa_config.sv"
```

```
 `include "sa_transaction.svh"
 `include "sa_sequencer.sv"
 `include "sa_monitor.sv"
 `include "sa_driver.sv"
 `include "sa_agent.sv"
 `include "sa_scoreboard.sv"
 `include "sa_env.sv"
 `include "sa_test.sv"
endpackage: sa_pkg
`endif /* __SA_PKG_SV__ */
```

sa_tb_top.sv

This is infrastructure source: trace each create/get/set/connect and identify where the interface handle is first bound.

Complete source: sa_tb_top.sv (22 lines)

```
`timescale 1ns/1ps
`include "sa_if.sv"
`include "sa_pkg.sv"
module sa_tb_top;
  import uvm_pkg::*;
  import sa_pkg::*;
  sa_if vif();
  sa dut(.clk(vif.sig_clock),.en_i(vif.sig_en_i),.ina(vif.sig_ina),
                 .inb(vif.sig_inb),.en_o(vif.sig_en_o),.result_o(vif.sig_out));
  initial begin
    sa_config cfg;
    cfg=sa_config::type_id::create("cfg");
    cfg.m_vif=vif;
    cfg.is_active_sa=UVM_ACTIVE;
    cfg.checks_enable_sa=1;
    cfg.coverage_enable_sa=1;
    uvm_config_db#(sa_config)::set(null,"uvm_test_top.sa_env","config",cfg);
    run_test("sa_test");
  end
  initial vif.sig_clock=0;
  always #5 vif.sig_clock=~vif.sig_clock;
endmodule
```

9. Compiling and debugging in Questa

Compile VHDL before the SystemVerilog top level. The supplied source includes simulator scripts and a DPI C++ predictor; make sure the C++ shared library is built and loaded with the correct architecture/toolchain before simulation.

```
set UVMF_HOME=C:\UVMF_2023.4
vlib work
vcom -2008 -work work sa.vhd
vlog -sv -work work +incdir+. sa_tb_top.sv
```

The commands above are illustrative; use the packaged compile_sa.do / sim_ver_sa.do as the canonical project-specific starting point. This is a conventional UVM project, not automatically a generated UVMF project.

compile_sa.do

Follow each command in order and check the expected compiled library and simulation top name.

Complete source: compile_sa.do (8 lines)

```
# Questa/ModelSim mixed-language compilation; run from UVM/ directory.
# Requires a Questa license supporting mixed-language VHDL + SystemVerilog.
if {![file exists work]} {vlib work}
# Compile the VHDL DUT first (VHDL-2008).
vcom -2008 -work work sa.vhd
# Compile the SystemVerilog/UVM testbench second.
vlog -sv -work work +incdir+. sa_tb_top.sv
# C++ DPI predictor requires a separate shared-library build.
```

sim_ver_sa.do

Follow each command in order and check the expected compiled library and simulation top name.

Complete source: sim_ver_sa.do (9 lines)

```
# Launch after building the DPI shared library and setting PREDICT_LIB.
do compile_sa.do
if {![info exists PREDICT_LIB]} {
  echo "ERROR: set PREDICT_LIB to the C++ DPI shared-library path (without extension)"
} else {
  vsim -c -sv_lib $PREDICT_LIB work.sa_tb_top +UVM_TESTNAME=sa_test
  run -all
  quit -sim
}
```

wave_dut.do

Follow each command in order and check the expected compiled library and simulation top name.

Complete source: wave_dut.do (2 lines)

```
add wave -divider "VIRTUAL IF"
add wave -position insertpoint sim:/sa_tb_top/vif/*
```

wave_transactions.do

Follow each command in order and check the expected compiled library and simulation top name.

Complete source: wave_transactions.do (3 lines)

```
# UVM transactions (paths depend on simulator/UVM configuration)
# Use Questa GUI transaction viewer to browse /uvm_root/uvm_test_top/sa_env/sa_agent
# Driver: sa_drvr; monitors: input_mon and output_mon
```